STAFF REVIEW · H446 095 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Rate of completed instructions.

2. Dependency/resource/control issue that interrupts pipeline.

3. Cache.

4. Q becomes0, sampled at the active edge.

I Do — source worked steps

1. Explain the mechanism — 2+3−1=4 cycles; serial baseline 2×3=6 cycles. — The second task starts stage one while first advances to stage two.

We Do — guided CHECK

1. Explain the mechanism
4+5−1=8 cycles, compared with 4×5=20 serial stage cycles.
Assumes equal one-cycle stages, no resource conflicts or stalls.

You Do — source mark schemes

1. I1 F1 D2 E3 [2], I2 F2 D3 E4 [2], I3 F3 D4 E5 [2].

2. Pipeline3+6−1=8 cycles [3], serial6×3=18 [2], equal stage/no stall assumption [1].

3. Data dependency waiting [2], branches flush/stall [2], average throughput drops though per-instruction latency not necessarily improved [2].

Exit Ticket — source answer

1. False. It overlaps stages, improving throughput when conditions allow.